DRUNKEN SAILOR DESIGN DOCUMENT

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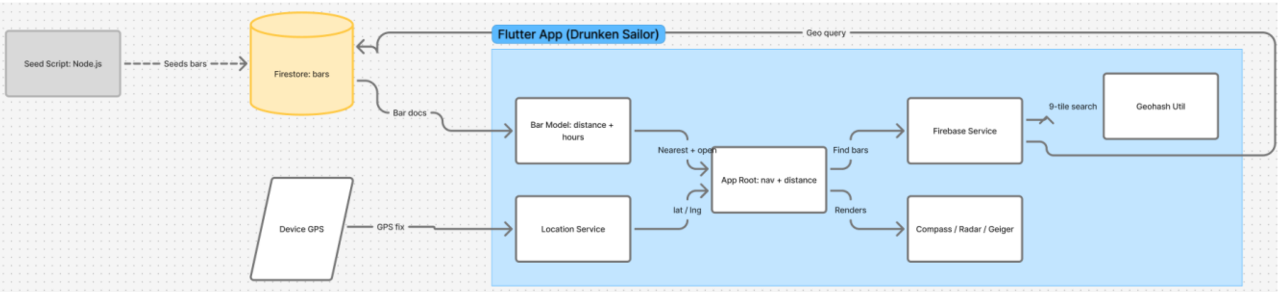
DRUNKEN SAILOR Group 1

Contenido

Diagrams

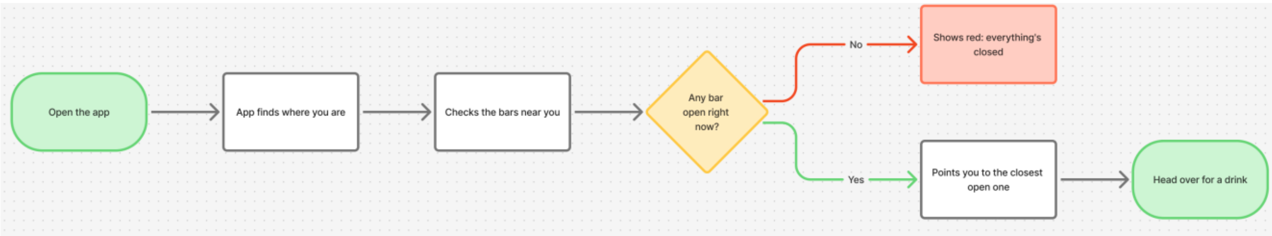
This chapter explains the main structure and behaviour of the Drunken Sailor app using different UML and process diagrams. The diagrams show how the app is built, how the user moves through the app, and which external services are needed, such as GPS, sensors, vibration, and Firebase.

Architecture Diagram



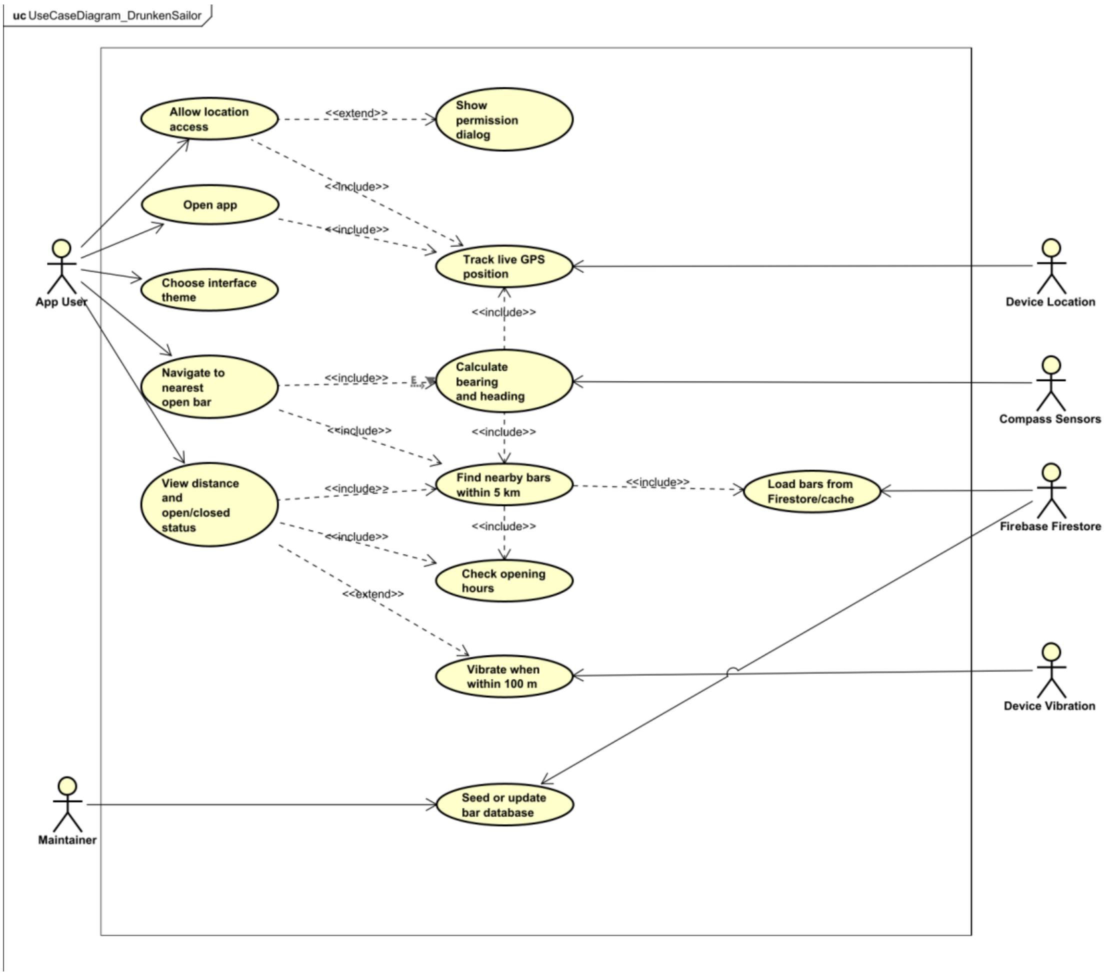
The architecture diagram shows the main technical parts of the application and how they communicate with each other. The user interacts with the Flutter interface, which contains the compass, radar, and Geiger counter views. These views use services for location, heading, vibration, and bar data. The bar data is loaded from Firebase Firestore and can also be cached so the app can still work better when the connection is unstable.

Flowchart Diagram



The flowchart diagram shows the main flow of the app from startup to finding a bar. First, the app starts and asks for location permission. If permission is accepted, the app gets the user's location and loads nearby bar data. The app then checks which bars are open and shows the nearest open bar. If no open bar is found, it shows the nearest closed bar instead.

Use Case Diagram



The use case diagram shows what the user can do with the app, and which external systems are involved. The main user can open the app, allow location access, choose between the compass, radar, and Geiger counter views, and navigate to the nearest open bar. The app also communicates with device services such as GPS, compass sensors, vibration, and Firebase Firestore. The maintainer can update or seed the bar database.

App Design Iterations

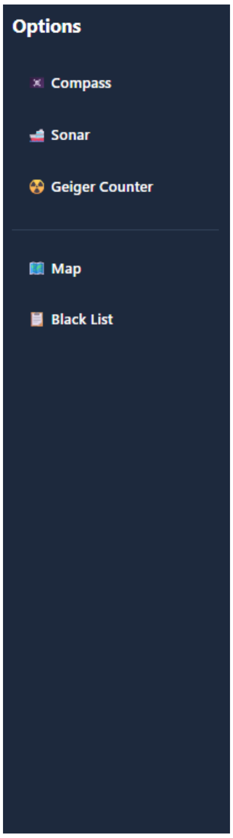
This chapter shows how the visual design of the app changed over time. Each iteration improved the interface by making it more themed, more readable, and more suitable for the final concept of the app. The designs started simple and became more polished with stronger visual styles for each mode.

Burger Menu

The burger menu gives the user a simple way to switch between the main app views. It contains the Compass, Radar, and Geiger Counter options, as well as planned options such as Map and Blacklist. This menu helps keep the main screen clean while still allowing navigation to the different app modes.

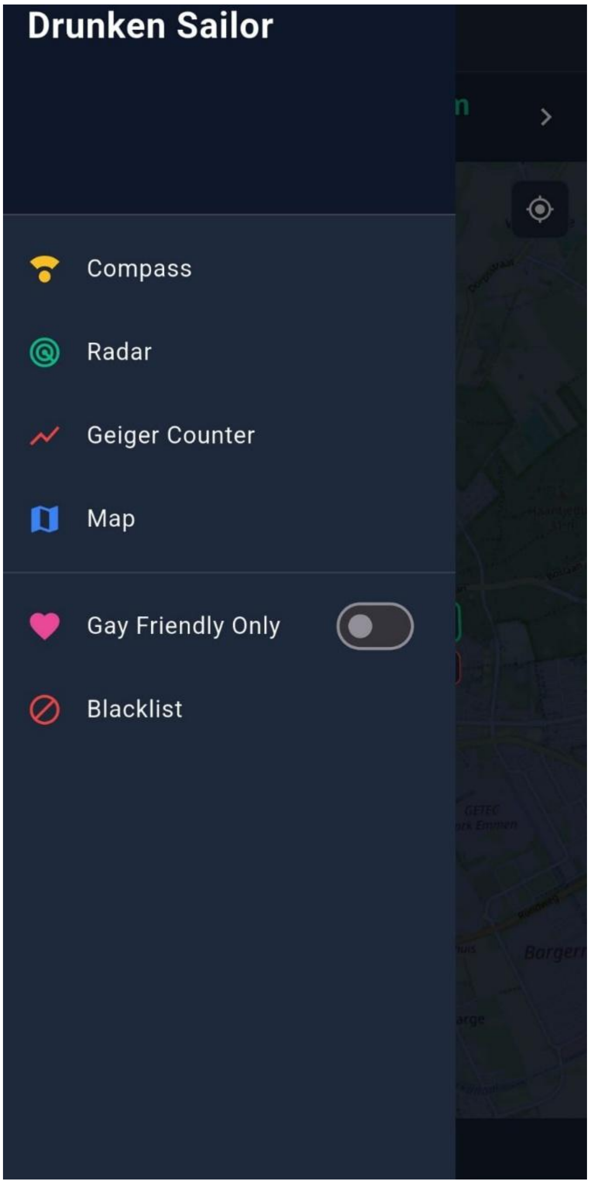
Iteration 1

The first burger menu iteration is a simple side menu design. It shows the main navigation options, including Compass, Sonar, Geiger Counter, Map, and Blacklist. The focus of this version was to create a basic structure for switching between the different app views.



Iteration 2

The second burger menu iteration improves the layout and makes it look more like a finished mobile app menu. It adds clearer icons, better spacing, a stronger dark theme, and extra options such as the gay-friendly filter. This version feels more polished and easier to use while keeping the navigation simple.

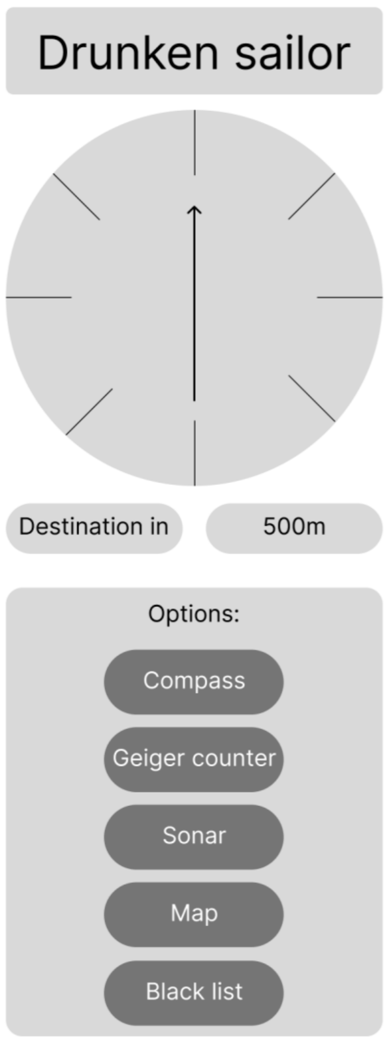


Compass

The compass view is designed to point the user toward the nearest open bar. It uses the phone's location and compass heading to calculate the direction of the target. The different iterations show how the compass changed from a simple wireframe into a more themed pirate-style interface.

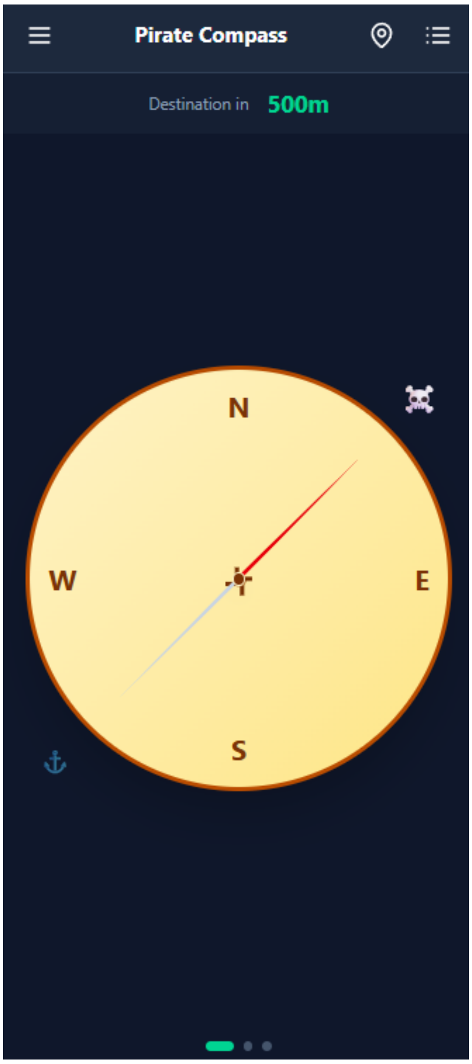
Iteration 1

The first compass iteration is a basic wireframe. It shows the main layout of the screen, including the compass, destination information, distance, and navigation buttons. The focus of this version was functionality and structure rather than visual design.



Iteration 2

The second compass iteration adds a darker mobile interface and begins to look more like the final app. It includes a top navigation bar, a distance indicator, and a compass in the centre of the screen. This version improves the layout and makes it feel more like an actual mobile application.

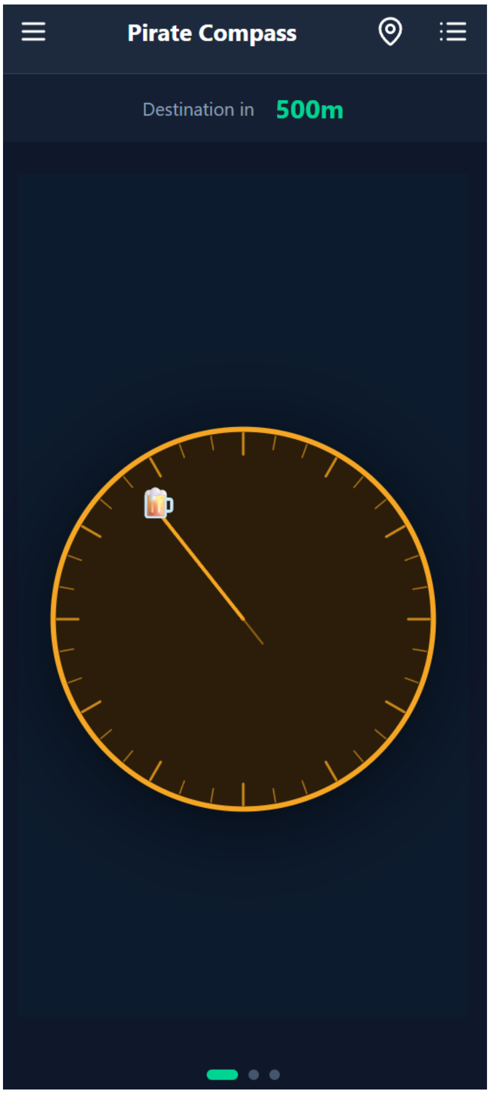


Iteration 3

The third compass iteration adds a pirate-inspired visual style. The wooden background gives the screen more personality and connects the design better to the “Drunken Sailor” theme. The compass remains the main focus of the screen.

Iteration 4

The fourth compass iteration simplifies and improves the final visual style. The compass becomes cleaner and easier to read, while still keeping the pirate theme through colours and details. This version feels more polished and usable than the earlier designs.

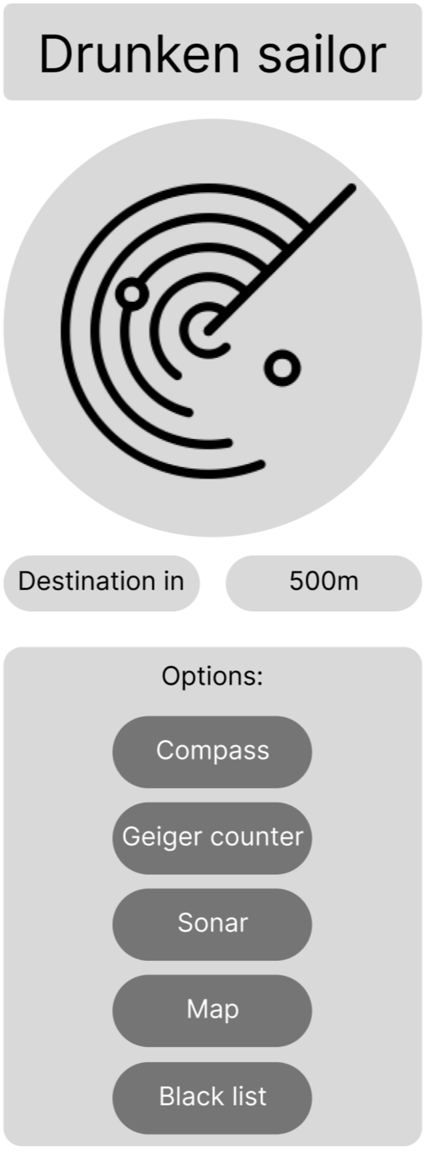


Radar

The radar view gives the user another way to see nearby bars. Instead of pointing directly like a compass, it shows bars as points on a radar screen. This fits the submarine theme and gives the user a more visual overview of nearby locations.

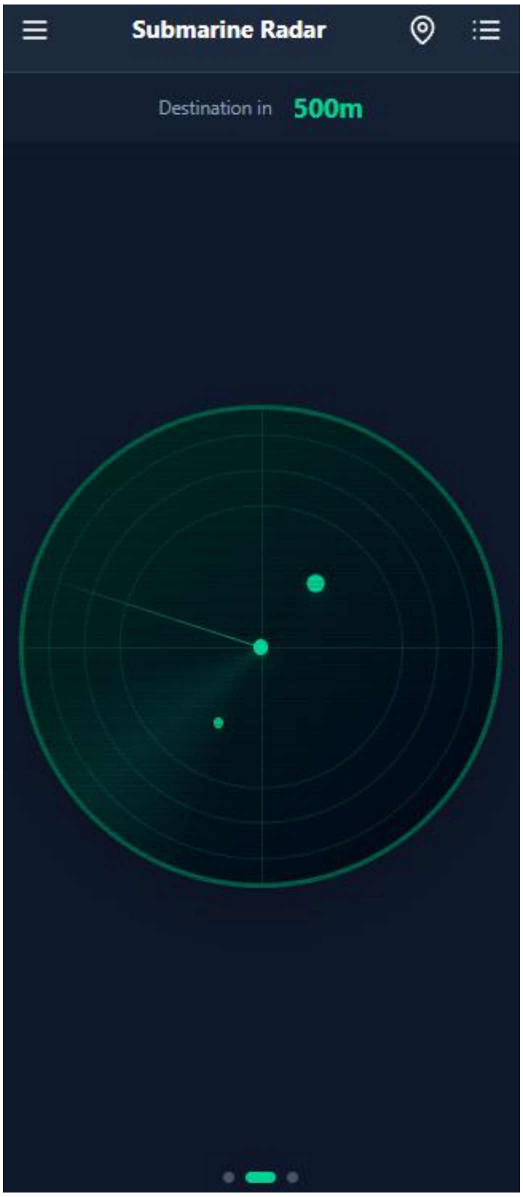
Iteration 1

The first radar iteration is a simple wireframe. It shows the basic idea of a radar display with nearby points and the same general navigation structure as the rest of the app. This version focuses on the concept rather than detailed styling.



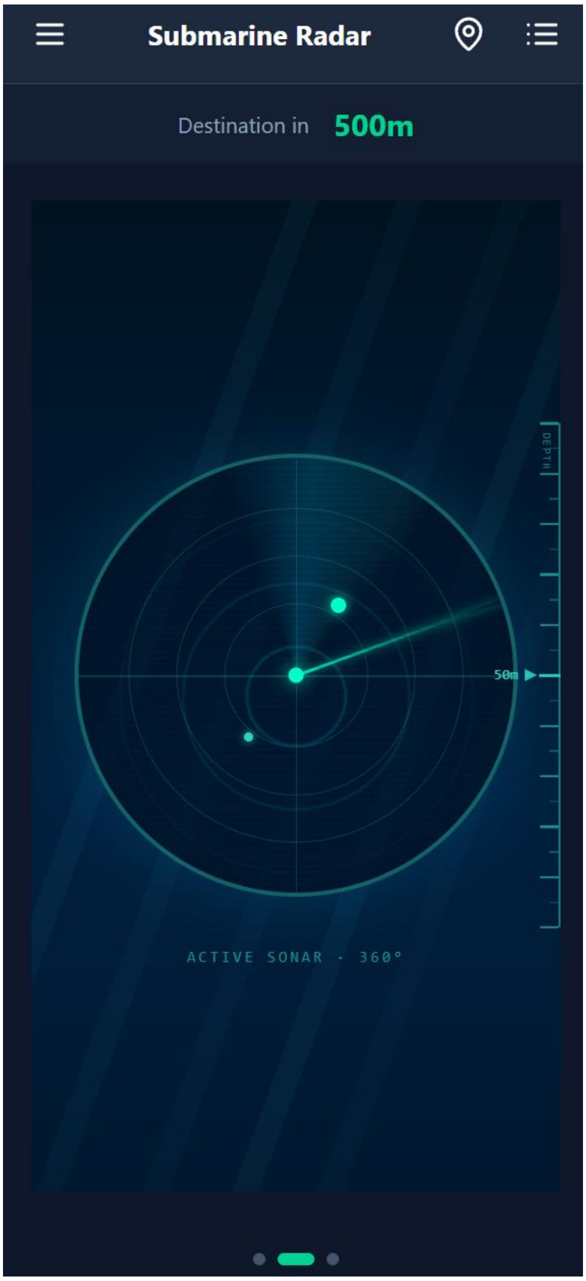
Iteration 2

The second radar iteration adds the darker app interface and green radar visuals. The bars are shown as dots on the radar, making it easier to understand their relative position. This version already matches the submarine theme more clearly.



Iteration 3

The third radar iteration improves the visual atmosphere with a stronger underwater style. The background and radar effects make the screen feel more immersive while keeping the radar readable. This version is closer to the final intended design.

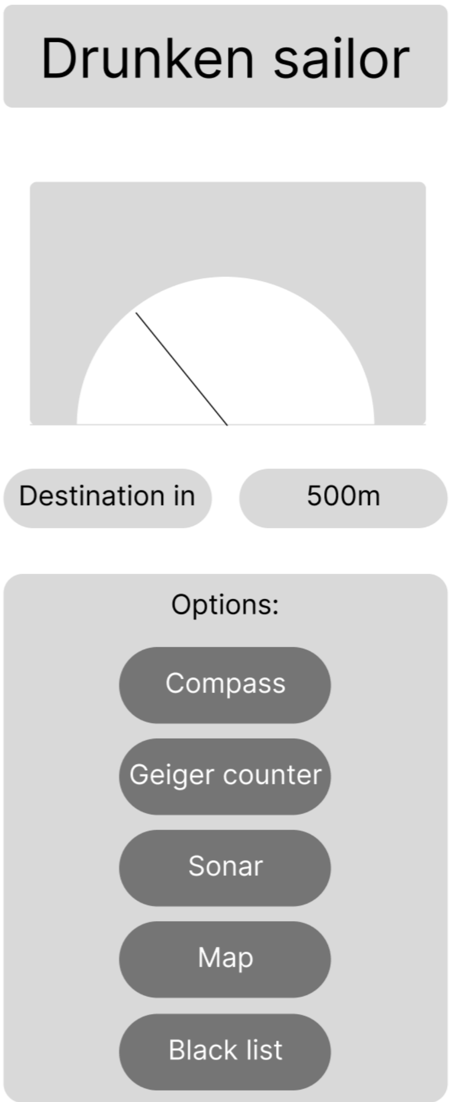


Geiger Counter

The Geiger counter view uses a nuclear/radiation theme to show how close the user is to a bar. Instead of only showing direction, it focuses on intensity and distance. The app can also use vibration when the user gets close to the target.

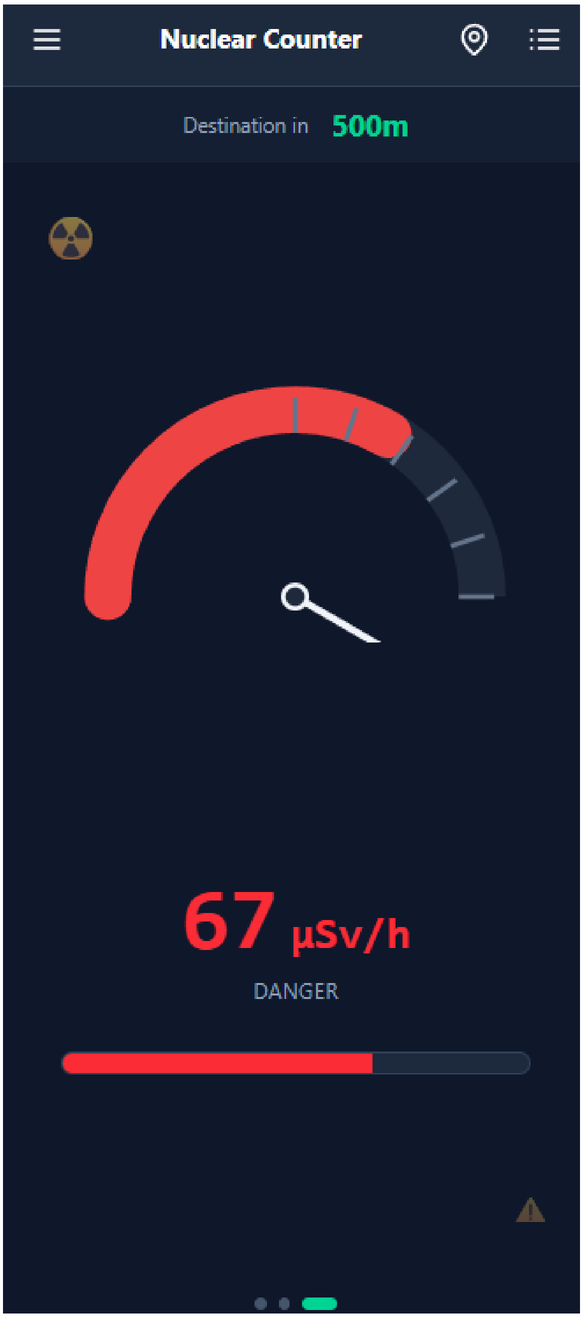
Iteration 1

The first Geiger counter iteration is a basic wireframe. It shows a simple meter, destination information, distance, and navigation buttons. The goal was to plan the layout before adding the nuclear visual theme.



Iteration 2

The second Geiger counter iteration introduces the dark interface and a radiation meter style. It includes a clear distance indicator and a strong red visual focus, which makes the view feel different from the compass and radar screens.

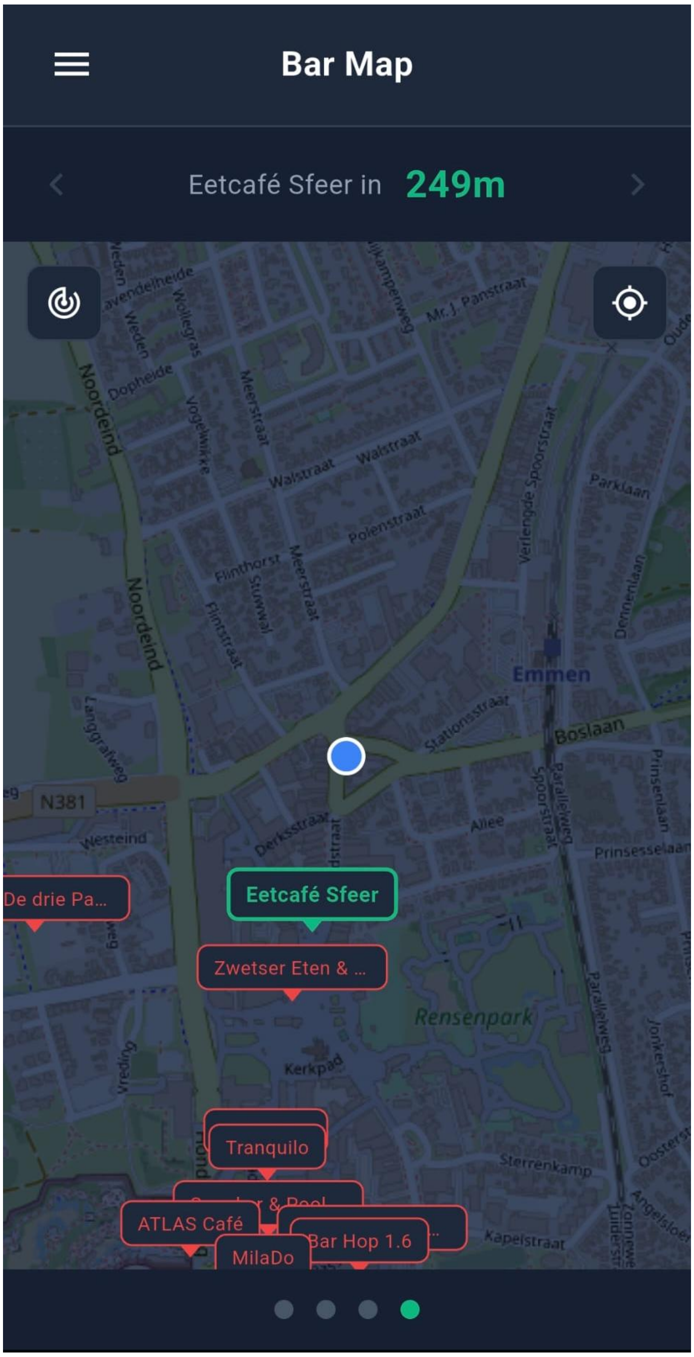


Iteration 3

The third Geiger counter iteration adds more nuclear-themed details, such as warning stripes and stronger colours. This makes the screen more distinctive and improves the connection between the design and the Geiger counter concept.

Map

The map offers a support for navigating streets to reach the destination as well as providing an overview of the closest bars.



Used Libraries and Components

The Drunken Sailor app is built with Flutter and Dart. Flutter is used to create the mobile user interface, handle navigation between screens, and build the themed visual designs for the Compass, Radar, Geiger Counter, and Map views. The app also uses Firebase Firestore to store and load bar data, and device services such as GPS, compass sensors, permissions, and vibration.

Used Libraries

Library	Purpose
Flutter	Main framework used to build the mobile application.
Firebase core	Initializes Firebase inside the Flutter app.
Cloud Firestore	Reads bar data from Firebase Firestore.
Firebase Analytics	Provides Firebase analytics support.
Flutter Riverpod	Manages app state and shares data between views and services.
Geolocator	Gets the user's GPS location and helps calculate distance to nearby bars.
Permission Handler	Requests and checks location permissions.
Sensors Plus	Accesses device sensor data for heading and compass behaviour.
Flutter Compass	Reads compass heading data from the device.
Vibration	Provides vibration feedback when the user gets close to a target.
Flutter Map	Displays the map view.
Latlong2	Handles latitude and longitude calculations for map and location features.
Shared Preferences	Stores small local data such as cached bar data or user settings.

Intl	Provides formatting and utility support.

Main App Components

Component	Description
Compass View	Drunken Pirate-themed screen that points the user toward the nearest open bar.
Radar View	Submarine-themed screen that shows nearby bars as radar targets.
Geiger Counter View	Nuclear-themed screen that shows closeness to a bar through intensity and distance.
Map View	Shows nearby bars on a map for easier street navigation.
Burger Menu / Navigation	Allows the user to switch between the different app views.
Location Service	Handles GPS updates and location accuracy.
Heading Service	Handles compass and sensor heading data.
Vibration Service	Handles phone vibration feedback.
Firestore Bar Repository	Loads bar data from Firebase Firestore.
Bar Cache	Stores bar data locally so the app works better when the connection is unstable.
Geohash Utility	Helps search for nearby bars efficiently based on location.
Location Permission Dialog	Shows themed messages when location permission is needed or denied.
Loading and Error Components	Show loading states and error messages when data or permissions are unavailable.